

taining security of remote factories is the skill-gap among embedded engineers. Most embedded engineers tend to be unfamiliar with cloud infrastructure and IT security concepts. Therefore they often struggle to create robust and secure Internet of Things (IoT) infrastructures without the support of IT experts.

Even a single weak link exposes the entire system to hackers, who typically seek just a single point of entry to obtain remote access to a large number of systems. There are several recent examples of large-scale damage caused by distributed denial of service (DDoS) attacks. Since the weakest link in an IoT network is generally the hardware, lack of IT knowledge among hardware engineers becomes even more problematic.

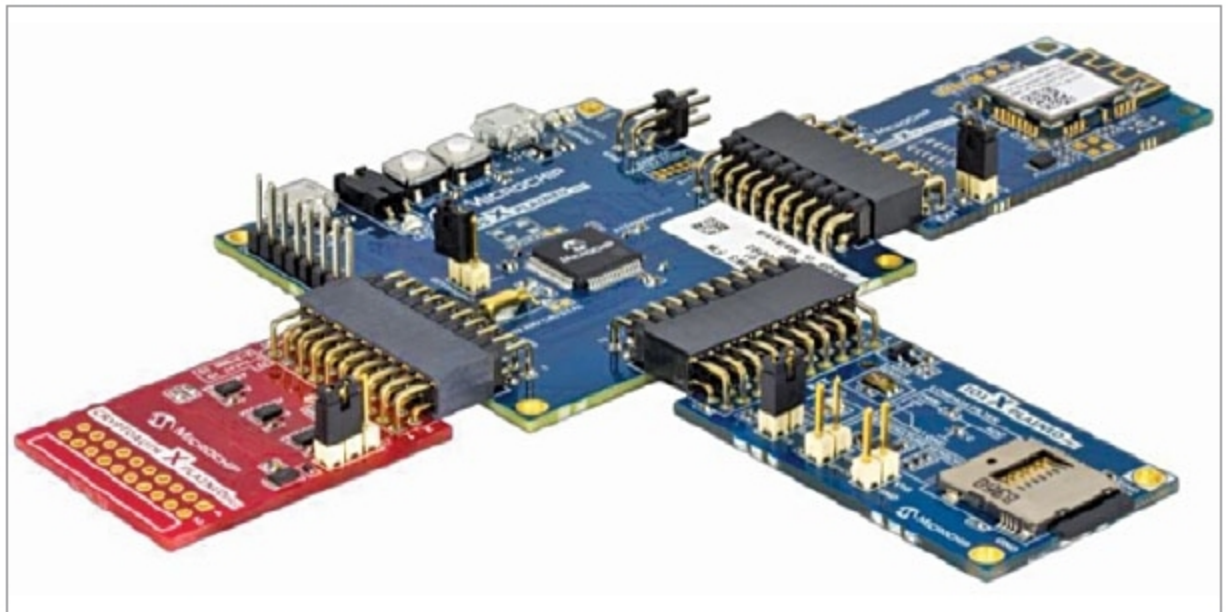
The industry is taking steps to address this issue. For instance, embedded chip companies, such as Microchip Technology, are doing their part to bridge this knowledge gap. They are working to educate engineers on steps to take to ensure end-to-end secure infrastructure.

Major cloud players such as Amazon Web Services (AWS), Google and Microsoft also have a role to play in sharing expertise, in addition to public key infrastructure (PKI) management companies such as Kyrio. Knowledge aside, it also requires a mindset change. Many a time, IoT implementations tend to consider the security aspect only after the entire network has been designed, which exposes it to security risks. Instead, it is important to take a strategic view of security, and incorporate it right at the hardware stage, rather than as an afterthought.

Authentication mechanisms

A flawless authentication mechanism is one of the most important components to consider while designing a secure system. Because every node is at risk of facing a security breach, it is critical that every node in the network has a unique identity that is protected, trusted and managed.

At all times, it is important to have visibility into who is on the



Microchip's CryptoAuthentication ICs

network. Are they who they say they are? Can they be trusted? Can they be monitored? To answer these questions, most systems traditionally use Transport Layer Security (TLS) 1.2 protocol. In this protocol, mutual authentication between a server and an IoT end node happens through a certificate authority—something that both sides trust.

There is, however, a downside to this. For this system to work, one needs to ensure protection of the trust issued by the certification authority all the way, right from the start of the project to manufacturing to the time the system is deployed. At the IoT end node, security and protection of the private key needs to be ensured, because this key is used to recognise the authenticity of the node.

The problem is that most organisations commonly store the private key within a microcontroller (MCU) in a flash memory. This practice is risky because the private key is vulnerable to software manipulations.

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This makes it possible for anyone to access and look at this memory zone, control it and then obtain the private key.

Even worse, presence of an authentication mechanism gives the designers of the network a false sense of security, which is the biggest danger that puts the system at risk.

There are two important measures that organisations need to take to ensure that the key is protected and secure. The first is to isolate the key and the second is to provision the chip with the key.

Isolating the key through a secure element. To ensure security of the key, it needs to be removed from the MCU. Not only that, the key and other critical credentials must be isolated from the MCU and from any potential software exposure.

One useful concept here is that of a secure element. The idea is to basically provide a safe haven for the key, where no one can access it. To validate authentication, all appropriate challenges/responses from the MCU need to be sent to the secure element based on commands from CryptoAuthLib library. The private key is never exposed, nor does it ever leave the secure element, whether during the product development process or the product lifecycle. This helps to ensure an end-to-end chain of trust.

Secure elements are essentially stand-alone CryptoAuthentication integrated circuits (ICs) that hold the private keys needed for IoT authentication. Conceptually, these